

Question	Type
What is your age?	Quantatative
Are you male or female?	Categorical
When did you first start reading the WSJ? High school, college, early career, midcareer, late career, or retirement?	Categorical
How long have you been in yout present job or position (in years)?	Quantatative
What type of vehivle are you considering for your next purchase? Nine response categories include sedan, sports car, SUV, minivan, and so on.	categorical

1. A Wall Street Journal subscriber survey asked 46 questions about subscriber characteristics and interests. State whether each of the questions to the left provides categorical or quantitative data.

Data	Bin Range	Frequency	Relative frequency	Percent frequency
14	12-14		2	0.05
19	15-17		8	0.2
24	18-20		11	0.275
19	21-23		10	0.25
16	24-26		9	0.225
20	total		40	1
24	12			
20	13		Percentiles:	
21	14		20th	16
22	15		25th	17.75
24	16		65th	22
18	17		75th	23
17	18			
23	19			
26	20			
22	21			
23	22			
25	23			
25	24			
19	25			
18	26			
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20				

2. Consider the Following Data.

A. Are these data categorical or quantitative?

Quantatitaive

B. Develop a frequency distribution using classes of 12-14, 15-17, 18-20, 21-23, and 24-26.

C. Develop a relative frequency distribution and a percent frequency distribution using the classes in part A.

D. Create a histogram graph using the bins from part A.

Histogram was not working, kept saying I had non numeric data in my bins, tried everything I could think of in the data analysis and numbers to try and fix it, just didn't work had everything done right

E. Compute the 20th, 25th, 65th, and 75th percentiles.

Year	Stivers (\$)	Trippi (\$)
1	11000	5600
2	12000	6300
3	13000	6900
4	14000	7600
5	15000	8500
6	16000	9200
7	17000	9900
8	18000	10600

3. Suppose that you initially invested \$10,000 in the Stivers mutual fund and \$5,000 in the Trippi mutual fund. The value of each investment at the end of each subsequent year is provided in the table.

A. Which of the two mutual funds performed better over this period? Why?
Trippi performed better due to a higher % return on investment

Salesperson	Total Sales (\$)	Average Performance Bonus Previous Years (\$)	Customer Accounts	Years with Company
Reeves, Bill	\$ 452,359.19	\$ 21,987.25	175	21
Smith, Michael	\$ 325,000.78	\$ 12,499.35	124	14
Graves, Lorrie	\$ 379,401.94	\$ 27,981.44	121	12
Quinn, Dorothy	\$ 234,091.39	\$ 14,567.98	48	9
Hamilton, Joshua	\$ 87,423.91	\$ 7,642.90	28	3
Harper, Derek	\$ 87,654.21	\$ 1,250.14	21	4
Thompson, Nicole	\$ 127,845.22	\$ 13,322.97	17	3
Yu, Joe	\$ 13,678.21	\$ 239.94	9	7
Sun, Yi	\$ 31,733.59	\$ 672.91	7	1

4. A sales manager is trying to determine appropriate sales performance bonuses for her team this year. The following table contains the data relevant to determining the bonuses, but it is not easy to read and interpret. Reformat the table to improve readability and to help the sales manager make her decisions about bonuses. List each step you make to reformat the table.

- 1. Used the excel dollar button to make average performance bonus more readable
- 2. Used excel dollar button on total sales to make it more readable and clearer
- 3. Highlighted those with over 10 years with company
- 4. Organized the chart to show customer accounts from most to least
- 5.
 - .
 - .
 - .
 - .
 - .(Add more if you need them)

Franchise	# U.S. Locations
Hampton Inns	1,864
ampm	3,183
McDonald's	32,805
7-Eleven, Inc.	37,496
Supercuts	2,130
Days Inn	1,877
Vanguard Cleaning Systems	2,155
Servpro	1,572
Subway	34,871
Denny's Inc.	1,668
Jan-Pro Franchising Intl Inc.	12,394
Hardee's	1,901
Pizza Hut Inc.	13,281
Kumon Math & Reading Centers	25,199
Dunkin' Donuts	9,947
KFC Corp.	16,224
Jazzercise Inc.	7,683
Anytime Fitness	1,618
Matco Tools	1,431
Stratus Building Solutions	5,018

5. Entrepreneur magazine ranks franchises. Among the factors that the magazine uses in its rankings are growth rate, number of locations, start-up costs, and financial stability. A recent ranking listed the top 20 U.S. franchises and the number of locations as in the table.

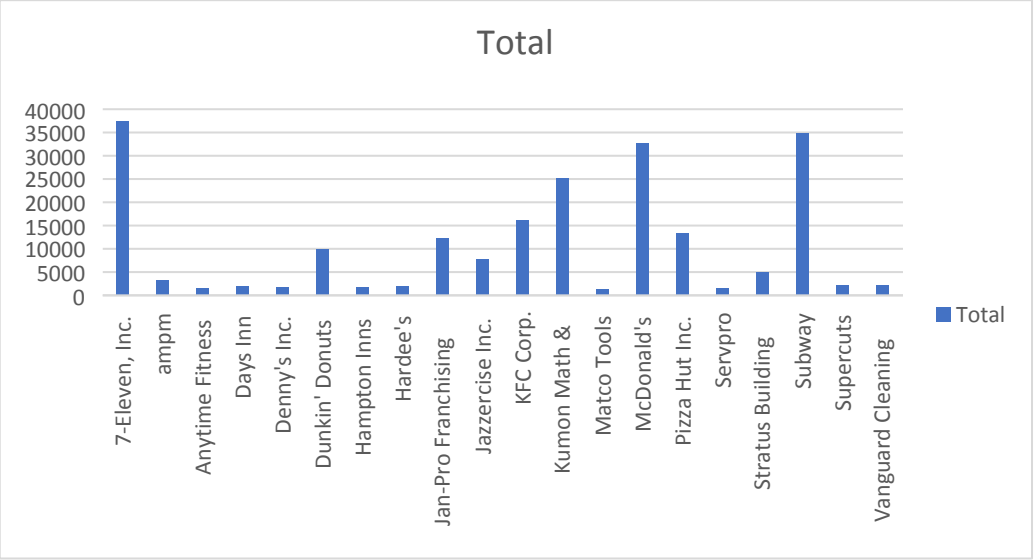
A. Create a PivotTable to summarize these data using classes 0-9,999, 10,000-19,999, 20,000-29,999, and 30,000-39,999 to answer the next questions. (Hint: Use Number of U.S. Locations as the COLUMNS, and use Count of Number of U.S. Locations as the VALUES in the PivotTable.

B. How many franchises have between 0 and 9,999 locations?

13

C. How many franchises have more than 30,000 locations?

3



Observation	x	y
1	-22	22
2	-33	49
3	2	8
4	29	-16
5	-13	10
6	21	-28
7	-13	27
8	-23	35
9	14	-5
10	3	-3
11	-37	48
12	34	-29
13	9	-18
14	-33	31
15	20	-16
16	-3	14
17	-15	18
18	12	17
19	-20	-11
20	-7	-22

6. The following table includes 20 observations for two quantitative variables x, and y.

A. Create a scatter chart for these 20 observations.

B. Fit a linear trendline to the 20 observation. What can you say about the relationship between the two quantitative variables?

The x and Y are negatively correlated as series 2 is going down as series 1 increases and vice versa

